PROGRAM:

AIES-Exp_3.py - C:/Users/ABBIRAMY/OneDrive/Documents/AIES_LAB/AIES-Exp_3.py (3.14.2)

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```
import heapq
graph = {
    'S': ['A', 'B'],
    'A': ['B', 'C', 'D'],
    'B': ['C'],
    'C': ['G'],
    'D': ['G'],
    'G': []
}
h = {'S': 7, 'A': 6, 'B': 4, 'C': 2, 'D': 3, 'G': 0}
def best_first(graph, h, start, goal):
    pq = [(h[start], start, [start])]   # (heuristic, node, path)
    visited = set()
    while pq:
        _, node, path = heapq.heappop(pq)
        if node == goal:
            return path
        if node in visited:
            continue
        visited.add(node)
        for nxt in graph[node]:
            if nxt not in visited:
                heapq.heappush(pq, (h[nxt], nxt, path + [nxt]))
    return None
print("Path:", best_first(graph, h, 'S', 'G'))
```

OUTPUT:

IDLE Shell 3.14.2

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```
Python 3.14.2 (tags/v3.14.2:df79316, Dec  5 2025, 17:18:21) [MSC v.1944 64 bit (
AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: C:/Users/ABBIRAMY/OneDrive/Documents/AIES_LAB/AIES-Exp_3.py =====
Path: ['S', 'B', 'C', 'G']
>>>
```